

## Cardiology [ACLS IN A NUTSHELL]

### Step 1: General Principles

The Step 2 exam will ask to either **identify the rhythm** or **choose an intervention**. In order to identify the rhythm, follow these simple principles. 1) Determine the **rate**: **tachycardia** is  $> 100$ , **bradycardia**  $< 60$ . 2) Determine the **QRS complex**: **wide** is  $> .12\text{msec}$  and means it's a **ventricular** rhythm while **narrow** is  $< .12\text{msec}$  and means it's an **atrial** rhythm. These two things will give you 80% of the answers on the test. The third and final decision is if the rhythm is **regular** or **irregular**. Of course, to determine any of this an ECG, preferably a 12-lead, is needed.

With the ECG ask if there's an arrhythmia or not. Note that there are two, maybe three, rhythms that aren't arrhythmias. **Normal Sinus Rhythm** is what everyone should be in. **Sinus tachycardia** is typically a normal, physiologic response to an underlying stressor. **Sinus bradycardia** may be a normal rhythm in a competitive athlete, though they usually do not appear in a vignette or in the hospital as an "arrhythmia."

### Step 2: Symptoms or No Symptoms

Ask, "are there symptoms?" An arrhythmia without any symptoms does not warrant attention. Simply: if there are **no symptoms** then **do nothing**. "Nothing" means routine care: IV, O<sub>2</sub>, and Monitor. Likely, this will be a question about rhythm identification.

### Step 3: Stable vs Unstable

If the patient has symptoms decide whether there's time to stay and play or if definitive therapy is needed right now. Stability is a product of your own comfort. But for a test, if there's **chest pain**, **shortness of breath**, **altered mental status**, or a **systolic BP  $< 90$** , then the patient is considered **unstable**. If they're unstable use **electricity**.

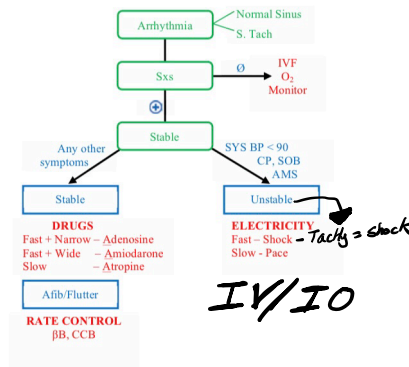
If instead the patient has symptoms, but not any one of those listed above, the patient is **stable**. A patient who is stable has time to fix the rhythm. They're not going to die at this moment; **pharmacotherapy** can be used.

### Step 4: Choose an intervention

If you've chosen **unstable/electricity** only one question needs to be asked - fast or slow. If the rhythm is **fast + unstable** then **shock**. If the rhythm is **slow + unstable** then **pace**.

If you've chosen **stable/electricity** it's a slightly more difficult task. For stable rhythms, there are three, maybe four, options. 1 - If the rhythm is **fast + narrow + stable** use **adenosine**. 2 - If the rhythm's **fast + wide + stable** use **amiodarone**. 3 - If the rhythm's **slow + stable** use **atropine** (epi drips can also be used in the new ACLS roll out). 4 - If the rhythm's **Afib/Aflutter** (note this is the only rhythm that actually had to be identified to do the right intervention), **rate control** is preferred. If they were unstable shock them since afib usually presents as a tachycardia. By "rate control" we mean **Beta Blockers** or **Calcium Channel Blockers**.

## OnlineMedEd



| Tachy Rhythms                                                                                                                                                                                                                          |                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>- Sinus Tachycardia</li><li>- Supraventricular Tachycardia</li><li>- Multifocal Atrial Tachycardia</li><li>- Afib</li><li>- Aflutter</li><li>- Vtach</li><li>- Vfib</li><li>- Torsades</li></ul> | <div><div>Atrial<br/>Narrow</div><div>Ventricular<br/>Wide</div></div> |
| Brady Rhythms                                                                                                                                                                                                                          |                                                                        |
| <ul style="list-style-type: none"><li>- Sinus Bradycardia</li><li>- 1° Block</li><li>- 2° Block</li><li>- 3° Block</li><li>- Junctional</li><li>- Idioventricular</li></ul>                                                            | <div><div>Varying degree<br/>of PR intervals</div></div>               |

| Intervention  | Heart Rate | QRS Complex  | Stability |
|---------------|------------|--------------|-----------|
| Pacer         | Brady      | Any          | Unstable  |
| Cardioversion | Tachy      | Any          | Unstable  |
| Atropine      | Brady      | Any          | Stable    |
| Adenosine     | Tachy      | Narrow       | Stable    |
| Amiodarone    | Tachy      | Wide         | Stable    |
| Rate Control  | Tachy      | Afib/Flutter | Stable    |

"Rate Control" = Verapamil / Diltiazem, Metoprolol

100-120 BPM  
CPR 30 → 2 Breaths  
2 in chest Rise  
2 min cycle pulse check → CPR  
switch CPR team

Target 32-36°C after ROA  
Temp 2.4 HR

Vent 1 Breath every 6sec

Brady - Atropine 1mg

DOP Infus 5-20mcg per minute

Reg narrow 50-100S

Pulse O<sub>2</sub> 92-98%

cardiac arrest  
epi 3-5min  
r-Fil r-tach  
1<sup>st</sup> ami 300mg  
2<sup>nd</sup> ami 150mg

Wave form capnography  
gold standard  
10min

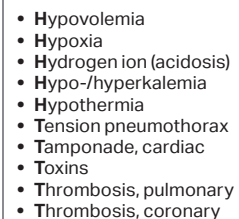
## 2020 Science Summary Table

This table compares 2015 with 2020, providing a quick reference to what has changed and what is new in the science of advanced cardiovascular life support.

**Table. Topical Comparison of 2015 and 2020 ACLS Science**

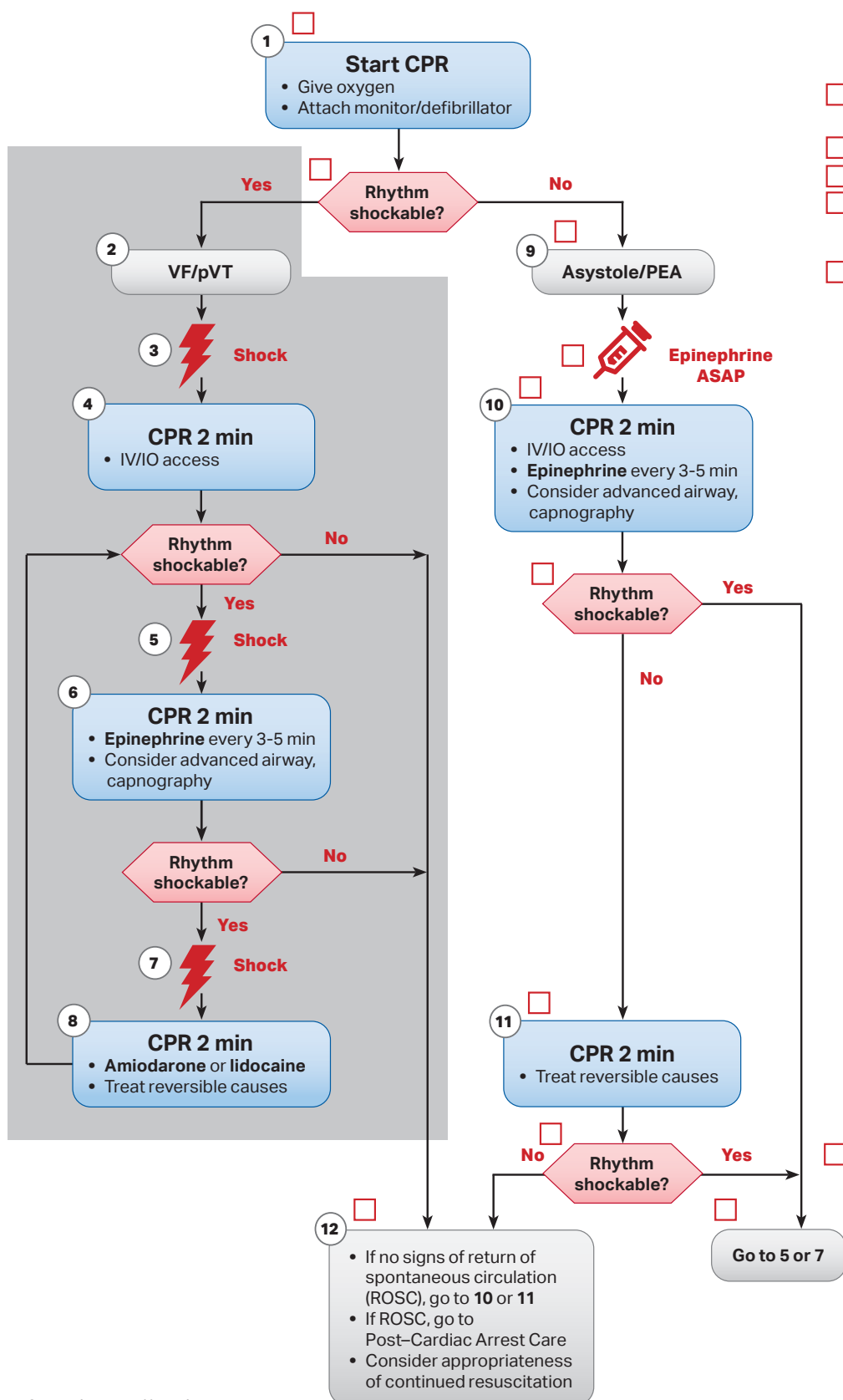
| ACLS topic                      | 2015                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2020                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ventilation</b>              | <ul style="list-style-type: none"> <li>1 breath every 5 to 6 seconds for respiratory arrest, with a bag-mask device</li> <li>1 breath every 6 seconds for ventilation with an advanced airway in place</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>1 breath every 6 seconds for respiratory arrest with or without an advanced airway and also for cardiac arrest with an advanced airway (use this rate with a bag-mask device if your local protocol is continuous compressions and asynchronous ventilations for cardiac arrest)</li> </ul> |
| <b>Bradycardia</b>              | <ul style="list-style-type: none"> <li>Atropine dose: 0.5 mg</li> <li>Dopamine dosing: 2 to 20 mcg/kg per minute</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>Atropine dose: 1 mg</li> <li>Dopamine dosing: 5 to 20 mcg/kg per minute</li> </ul>                                                                                                                                                                                                          |
| <b>Tachycardia</b>              | <ul style="list-style-type: none"> <li>Synchronized cardioversion initial recommended doses: <ul style="list-style-type: none"> <li>Narrow QRS complex, regular rhythm: 50 to 100 J</li> <li>Narrow QRS complex, irregular rhythm: 120 to 200 J</li> <li>Wide QRS complex, regular rhythm: 100 J</li> </ul> </li> <li>Wide QRS complex, irregular rhythm: defibrillation dose (not synchronized)</li> </ul>                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>Follow your specific device's recommended energy level to maximize the success of the first shock</li> <li>Wide QRS complex, irregular rhythm: defibrillation dose (not synchronized)</li> </ul>                                                                                            |
| <b>Post-Cardiac Arrest Care</b> | <ul style="list-style-type: none"> <li>Titrate oxygen saturation to 94% or higher</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Titrate oxygen saturation to 92% to 98%</li> </ul>                                                                                                                                                                                                                                          |
| <b>Adult Chain of Survival</b>  | <ul style="list-style-type: none"> <li>5 links for both chains (in-hospital cardiac arrest and out-of-hospital cardiac arrest)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>6 links for both chains (in-hospital cardiac arrest and out-of-hospital cardiac arrest): added a Recovery link to the end of both chains</li> </ul>                                                                                                                                         |
| <b>IV/IO Access</b>             | <ul style="list-style-type: none"> <li>IV access and IO access are equivalent</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>IV preferred over IO access, unless IV fails (then OK to proceed to IO)</li> </ul>                                                                                                                                                                                                          |
| <b>ACLS topic</b>               | <b>2020</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                    |
| <b>Cardiac Arrest</b>           | <ul style="list-style-type: none"> <li>Epinephrine 1 mg every 3 to 5 minutes or every 4 minutes as a midrange (ie, every other 2-minute rhythm check)</li> <li>Amiodarone and lidocaine are equivalent for treatment (ie, either may be used)</li> <li>Added maternal cardiac arrest information and algorithms (in-hospital)</li> <li>Added ventricular assist device information (left and right ventricular assist device) and algorithm</li> <li>Added new prognostication diagram and information</li> <li>Recommend using waveform capnography with a bag-mask device</li> </ul>                                                                                                    |                                                                                                                                                                                                                                                                                                                                    |
| <b>Stroke</b>                   | <ul style="list-style-type: none"> <li>Revised stroke algorithm</li> <li>New stroke triage algorithm for EMS destination</li> <li>Focus on large vessel occlusion for all healthcare providers</li> <li>Endovascular therapy: treatment window up to 24 hours (previously up to 6 hours)</li> <li>Both alteplase and endovascular therapy can be given/performed if time criteria and inclusion criteria are met</li> <li>Consider having EMS bypass the emergency department and go straight to the imaging suite (computed tomography [CT]/magnetic resonance imaging); initial assessment can be performed there to save time</li> <li>Titrate oxygen saturation to &gt;94%</li> </ul> |                                                                                                                                                                                                                                                                                                                                    |

## Adult Cardiac Arrest Algorithm (VF/pVT)



# Adult Cardiac Arrest Learning Station Checklist (Asystole/PEA)

## Adult Cardiac Arrest Algorithm (Asystole/PEA)



### CPR Quality

- Push hard (at least 2 inches [5 cm]) and fast (100-120/min) and allow complete chest recoil.
- Minimize interruptions in compressions.
- Avoid excessive ventilation.
- Change compressor every 2 minutes, or sooner if fatigued.
- If no advanced airway, 30:2 compression-ventilation ratio.
- Quantitative waveform capnography
  - If PETCO<sub>2</sub> is low or decreasing, reassess CPR quality.

### Shock Energy for Defibrillation

- **Biphasic:** Manufacturer recommendation (eg, initial dose of 120-200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered.
- **Monophasic:** 360 J

### Drug Therapy

- **Epinephrine IV/IO dose:** 1 mg every 3-5 minutes
- **Amiodarone IV/IO dose:** First dose: 300 mg bolus. Second dose: 150 mg.
- **Lidocaine IV/IO dose:** First dose: 1-1.5 mg/kg. Second dose: 0.5-0.75 mg/kg.

### Advanced Airway

- Endotracheal intubation or supraglottic advanced airway
- Waveform capnography or capnometry to confirm and monitor ET tube placement
- Once advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions

### Return of Spontaneous Circulation (ROSC)

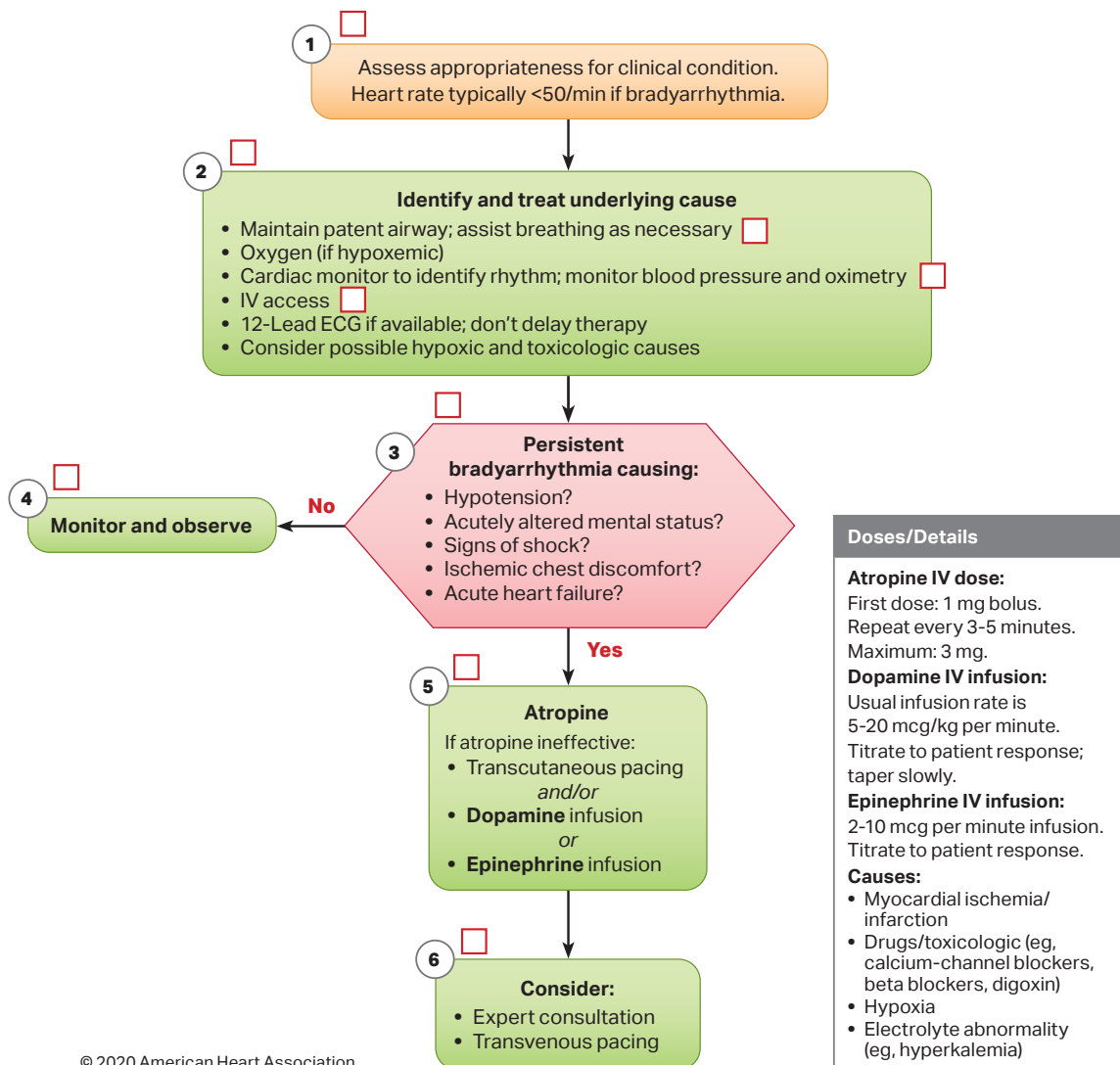
- Pulse and blood pressure
- Abrupt sustained increase in PETCO<sub>2</sub> (typically ≥40 mm Hg)
- Spontaneous arterial pressure waves with intra-arterial monitoring

### Reversible Causes

- Hypovolemia
- Hypoxia
- Hydrogen ion (acidosis)
- Hypo-/hyperkalemia
- Hypothermia
- Tension pneumothorax
- Tamponade, cardiac
- Toxins
- Thrombosis, pulmonary
- Thrombosis, coronary

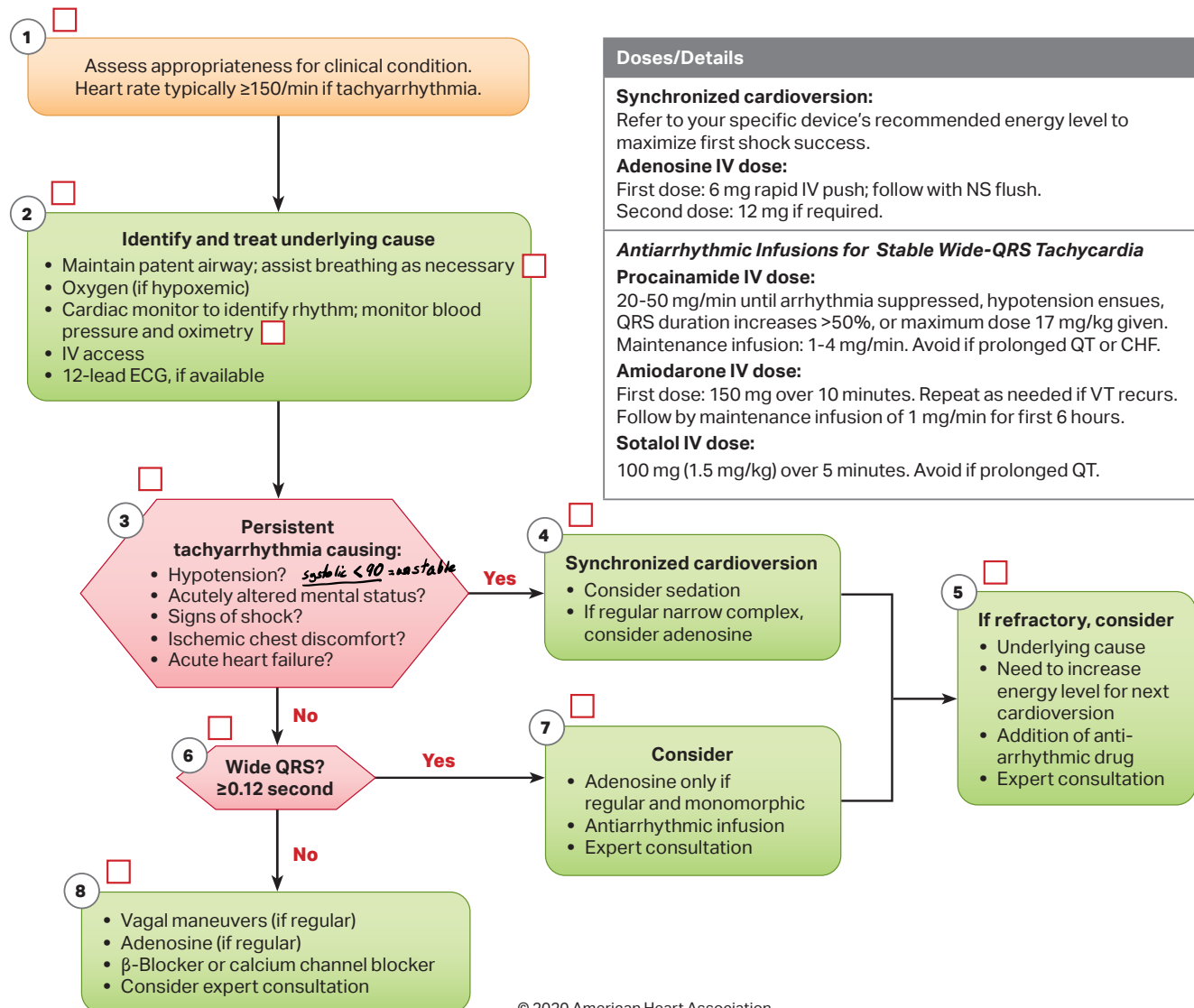
# Adult Bradycardia Learning Station Checklist

## Adult Bradycardia Algorithm



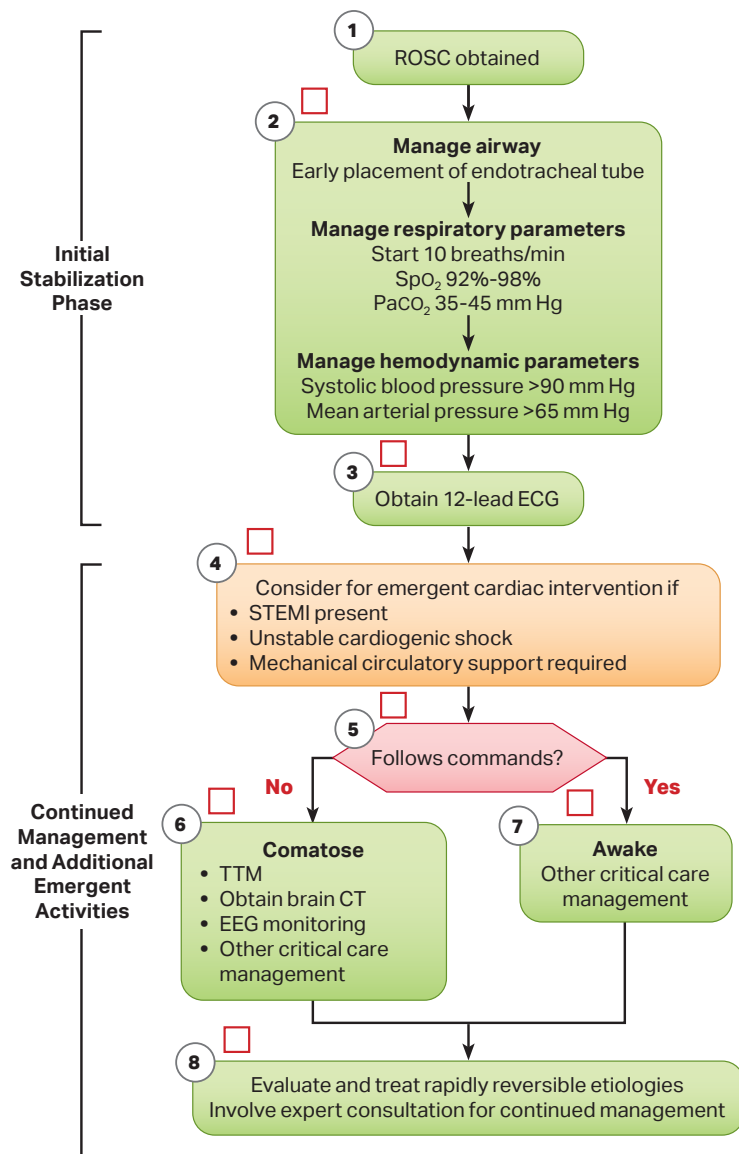
# Adult Tachycardia With a Pulse Learning Station Checklist

## Adult Tachycardia With a Pulse Algorithm



# Adult Post-Cardiac Arrest Care Learning Station Checklist

## Adult Post-Cardiac Arrest Care Algorithm



### Initial Stabilization Phase

Resuscitation is ongoing during the post-ROSC phase, and many of these activities can occur concurrently. However, if prioritization is necessary, follow these steps:

- **Airway management:**  
Waveform capnography or capnometry to confirm and monitor endotracheal tube placement
- **Manage respiratory parameters:**  
Titrate  $\text{FiO}_2$  for  $\text{SpO}_2$  92%-98%; start at 10 breaths/min; titrate to  $\text{PaCO}_2$  of 35-45 mm Hg
- **Manage hemodynamic parameters:**  
Administer crystalloid and/or vasopressor or inotrope for goal systolic blood pressure >90 mm Hg or mean arterial pressure >65 mm Hg

### Continued Management and Additional Emergent Activities

These evaluations should be done concurrently so that decisions on targeted temperature management (TTM) receive high priority as cardiac interventions.

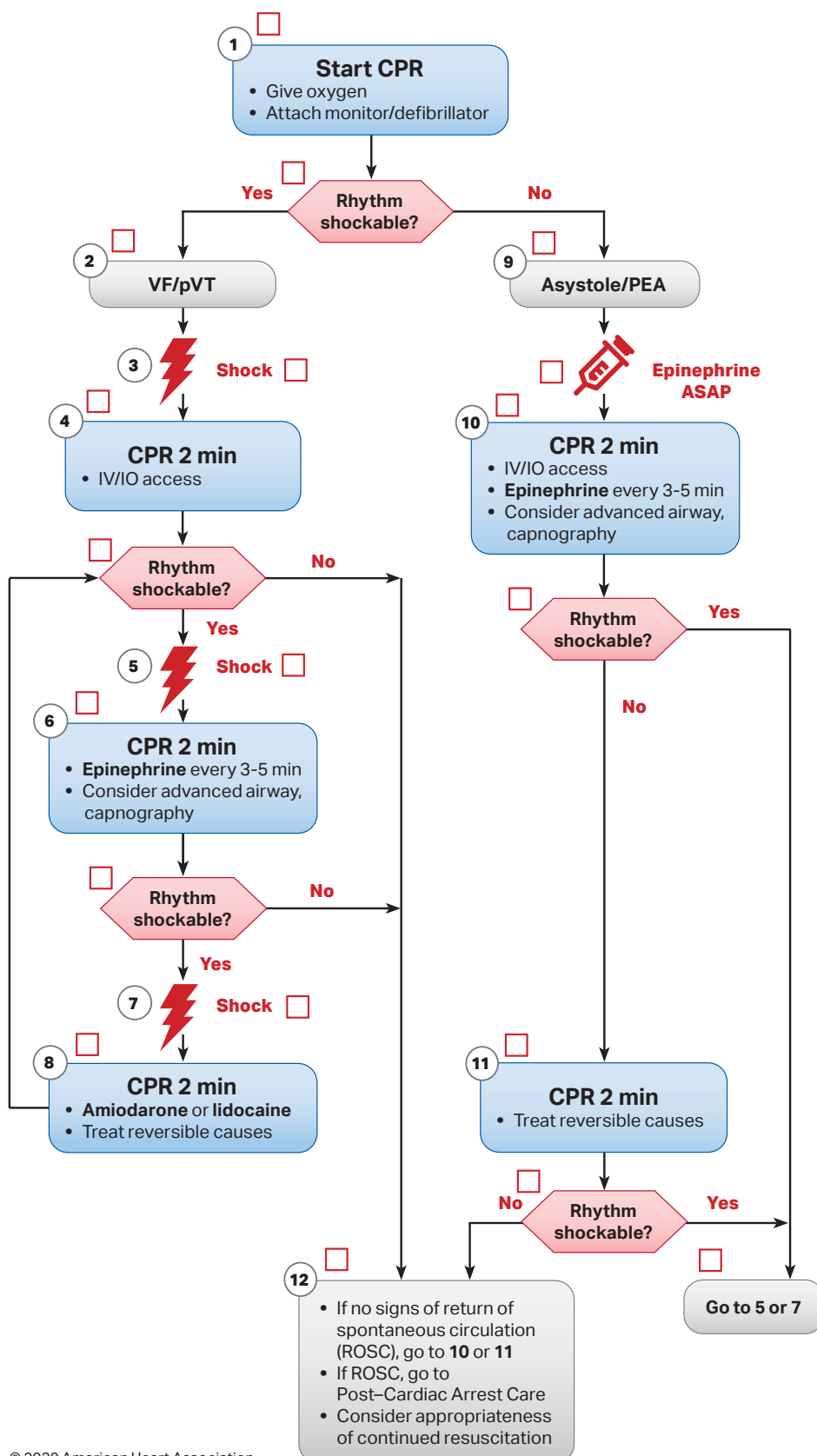
- **Emergent cardiac intervention:**  
Early evaluation of 12-lead electrocardiogram (ECG); consider hemodynamics for decision on cardiac intervention
- **TTM:** If patient is not following commands, start TTM as soon as possible; begin at 32-36°C for 24 hours by using a cooling device with feedback loop
- **Other critical care management**
  - Continuously monitor core temperature (esophageal, rectal, bladder)
  - Maintain normoxia, normocapnia, euglycemia
  - Provide continuous or intermittent electroencephalogram (EEG) monitoring
  - Provide lung-protective ventilation

### H's and T's

Hypovolemia  
Hypoxia  
Hydrogen ion (acidosis)  
Hypokalemia/hyperkalemia  
Hypothermia  
Tension pneumothorax  
Tamponade, cardiac  
Toxins  
Thrombosis, pulmonary  
Thrombosis, coronary

# Adult Cardiac Arrest Learning Station Checklist (VF/pVT/Asystole/PEA)

## Adult Cardiac Arrest Algorithm (VF/pVT/Asystole/PEA)



### CPR Quality

- Push hard (at least 2 inches [5 cm]) and fast (100-120/min) and allow complete chest recoil.
- Minimize interruptions in compressions.
- Avoid excessive ventilation.
- Change compressor every 2 minutes, or sooner if fatigued.
- If no advanced airway, 30:2 compression-ventilation ratio.
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  - If PETCO<sub>2</sub> is low or decreasing, reassess CPR quality.

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- **Epinephrine IV/IO dose:** 1 mg every 3-5 minutes
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- or
- **Lidocaine IV/IO dose:** First dose: 1-1.5 mg/kg. Second dose: 0.5-0.75 mg/kg.

### Advanced Airway

- Endotracheal intubation or supraglottic advanced airway
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### Return of Spontaneous Circulation (ROSC)

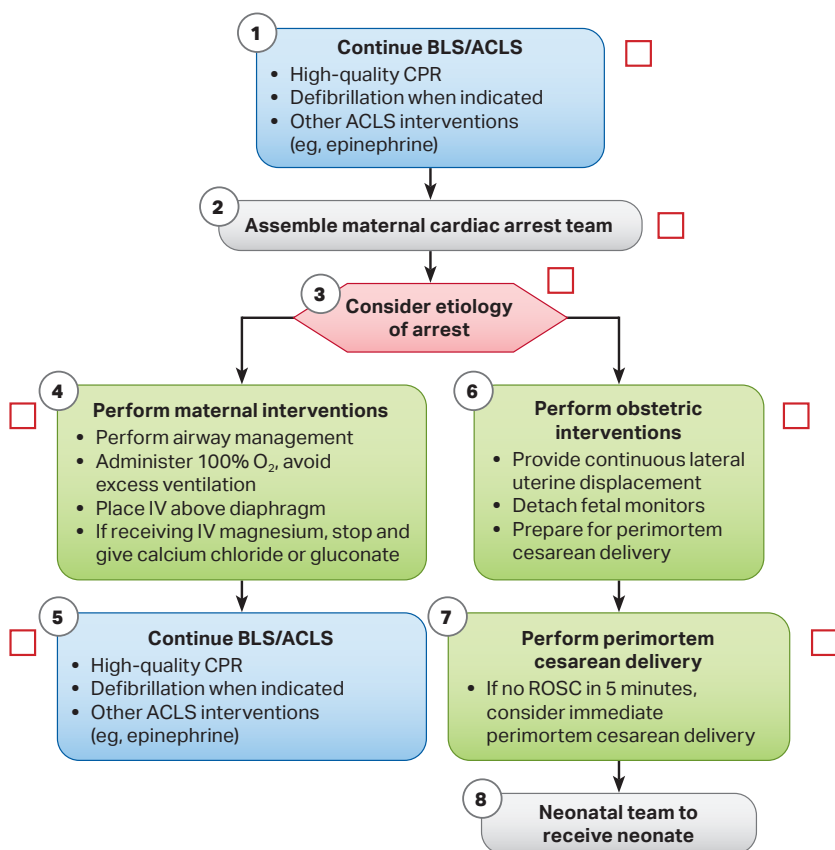
- Pulse and blood pressure
- Abrupt sustained increase in PETCO<sub>2</sub> (typically ≥40 mm Hg)
- Spontaneous arterial pressure waves with intra-arterial monitoring

### Reversible Causes

- Hypovolemia
- Hypoxia
- Hydrogen ion (acidosis)
- Hypo-/hyperkalemia
- Hypothermia
- Tension pneumothorax
- Tamponade, cardiac
- Toxins
- Thrombosis, pulmonary
- Thrombosis, coronary

# Cardiac Arrest in Pregnancy In-Hospital ACLS Learning Station Checklist

## Cardiac Arrest in Pregnancy In-Hospital ACLS Algorithm



### Maternal Cardiac Arrest

- Team planning should be done in collaboration with the obstetric, neonatal, emergency, anesthesiology, intensive care, and cardiac arrest services.
- Priorities for pregnant women in cardiac arrest should include provision of high-quality CPR and relief of aortocaval compression with lateral uterine displacement.
- The goal of perimortem cesarean delivery is to improve maternal and fetal outcomes.
- Ideally, perform perimortem cesarean delivery in 5 minutes, depending on provider resources and skill sets.

### Advanced Airway

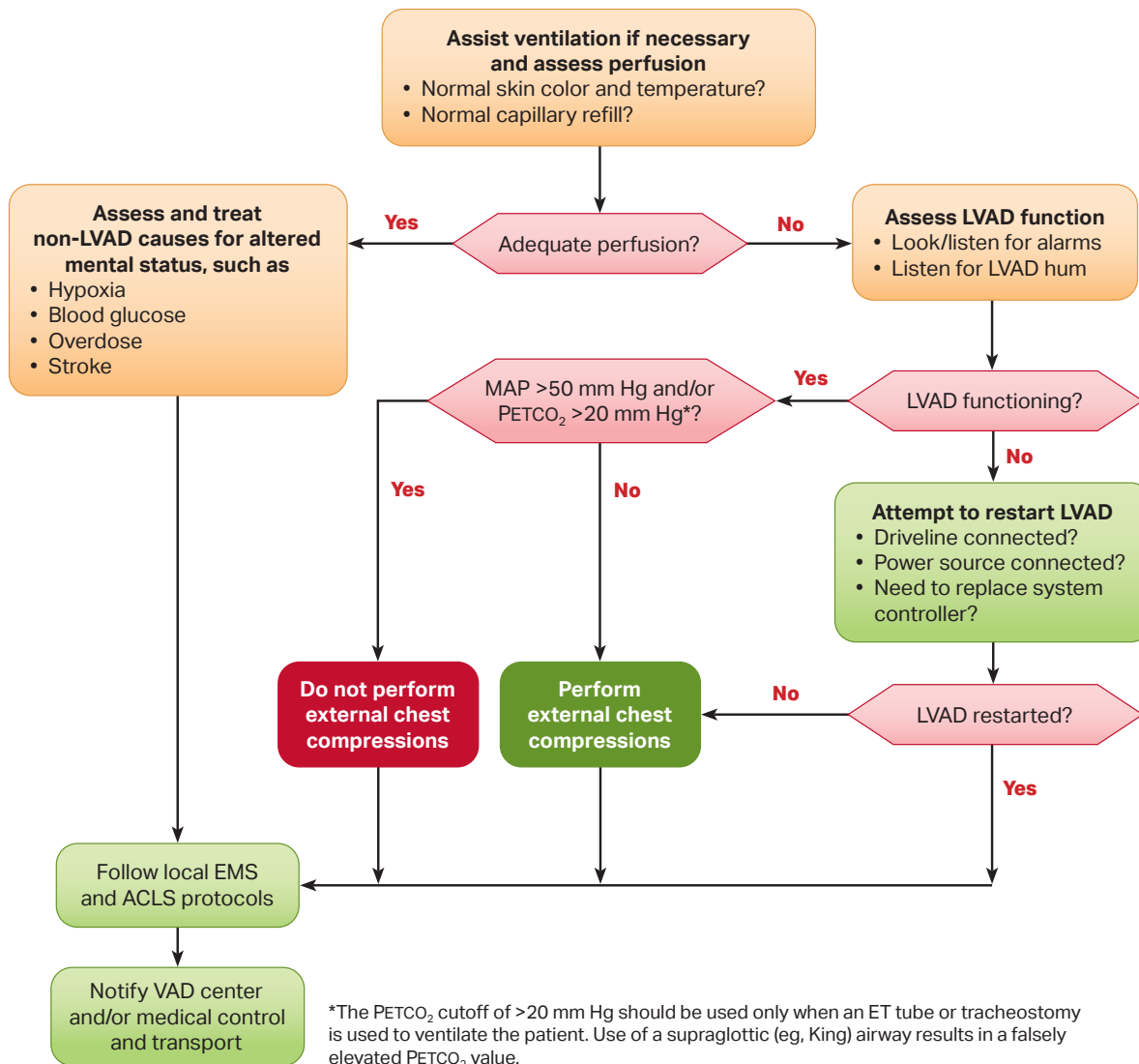
- In pregnancy, a difficult airway is common. Use the most experienced provider.
- Provide endotracheal intubation or supraglottic advanced airway.
- Perform waveform capnography or capnometry to confirm and monitor ET tube placement.
- Once advanced airway is in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions.

### Potential Etiology of Maternal Cardiac Arrest

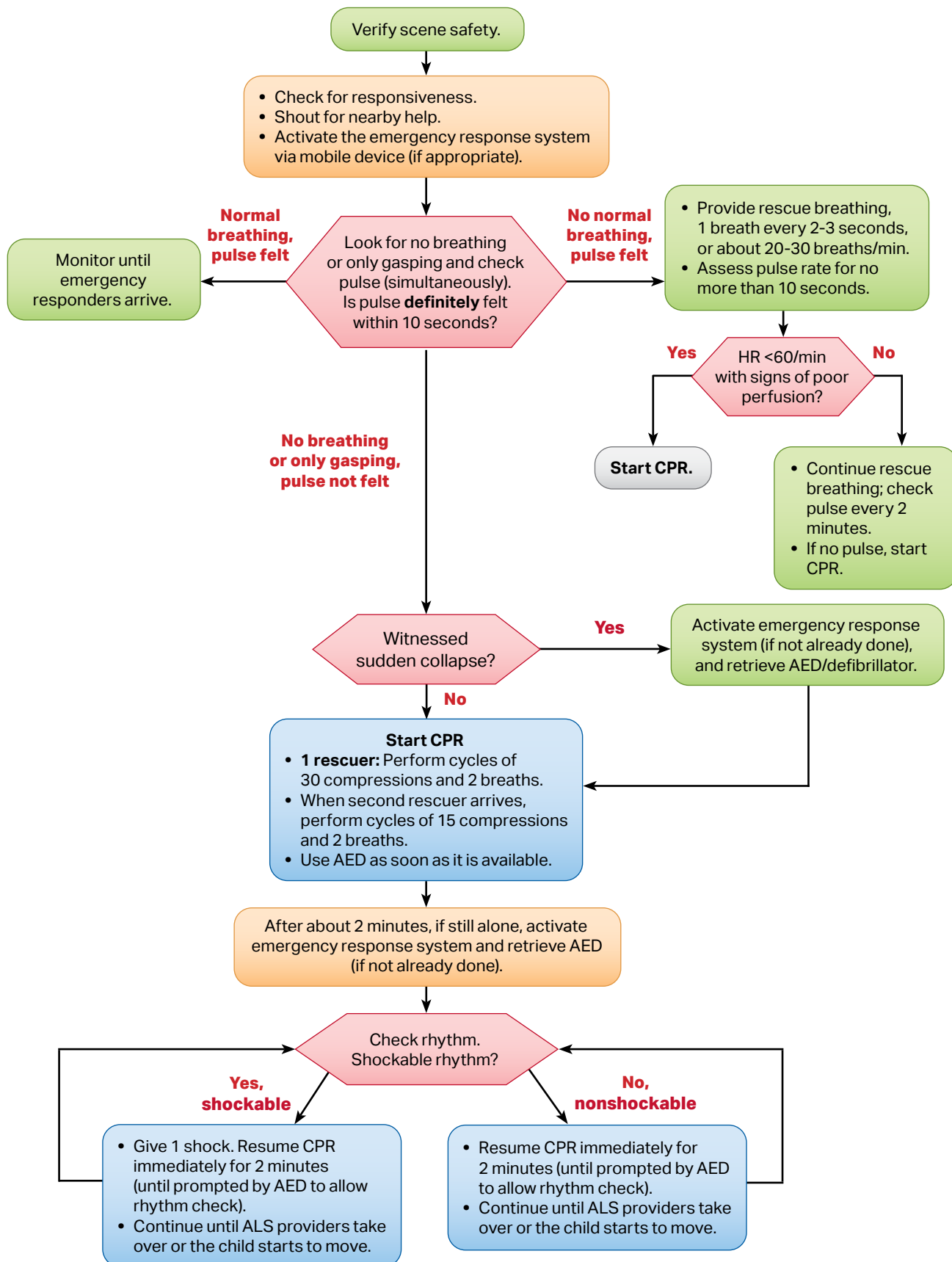
- A** Anesthetic complications
- B** Bleeding
- C** Cardiovascular
- D** Drugs
- E** Embolic
- F** Fever
- G** General nonobstetric causes of cardiac arrest (H's and T's)
- H** Hypertension

# Adult Ventricular Assist Device Learning Station Checklist

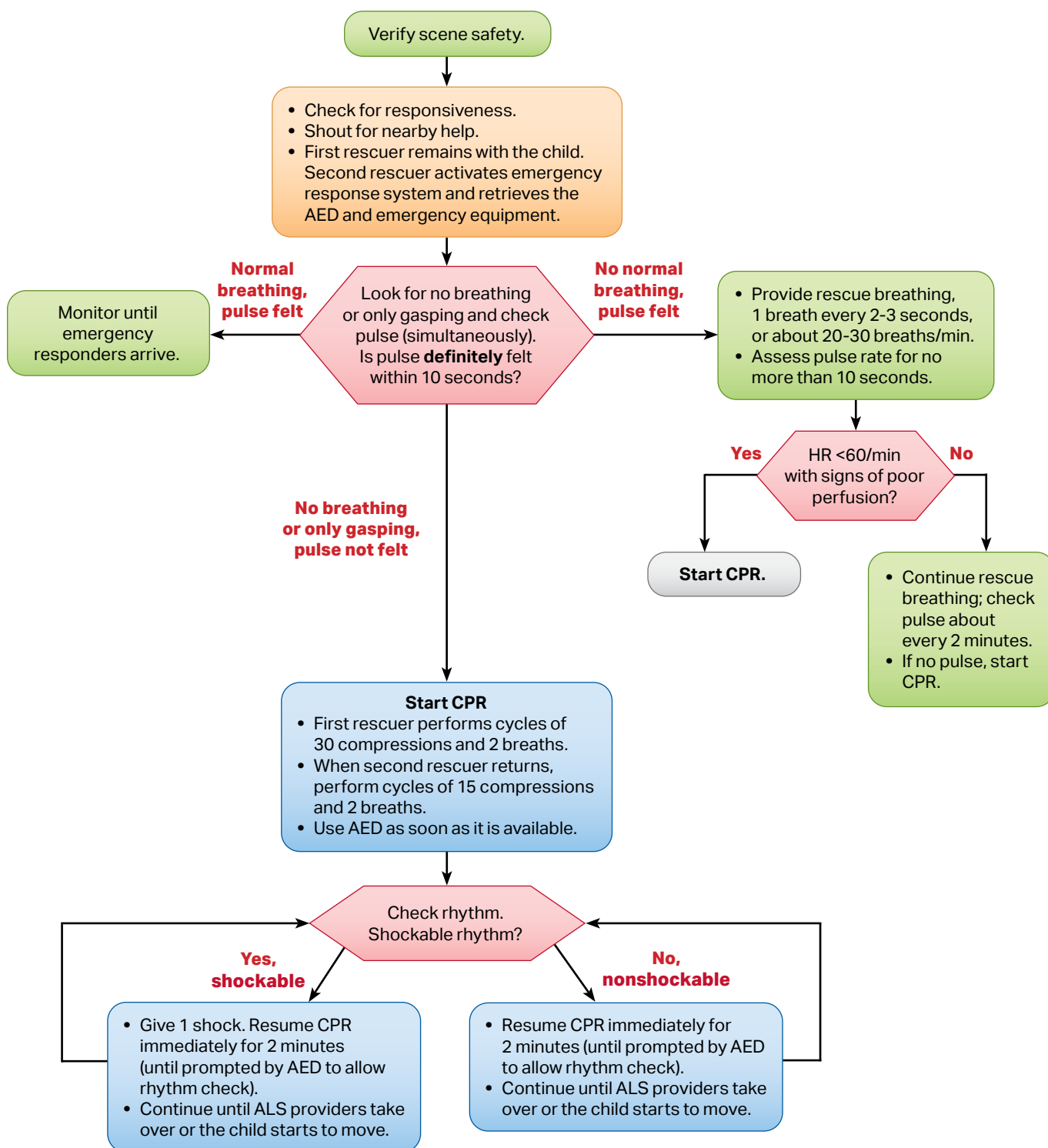
## Adult Ventricular Assist Device Algorithm



## Pediatric Basic Life Support Algorithm for Healthcare Providers—Single Rescuer



## Pediatric Basic Life Support Algorithm for Healthcare Providers—2 or More Rescuers



# Megacode Practice Learning Station Checklist: Case 48

## Tachycardia → VF → Asystole → PCAC

Student Name \_\_\_\_\_ Date of Test \_\_\_\_\_

| Critical Performance Steps                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    | Check if done correctly |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|-------------------------|
| <b>Team Leader</b>                                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Assigns team member roles                                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures high-quality CPR at all times                                                                                                                                               | Compression rate 100-120/min<br><input type="checkbox"/> | Compression depth of ≥2 inches<br><input type="checkbox"/> | Chest compression fraction >80%<br><input type="checkbox"/> | Chest recoil (optional)<br><input type="checkbox"/> | Ventilation (optional)<br><input type="checkbox"/> |                         |
| Ensures that team members communicate well                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Tachycardia Management</b>                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Starts oxygen if needed, places monitor, starts IV                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Places monitor leads in proper position                                                                                                                                             |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes unstable tachycardia                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes symptoms due to respiratory arrest (choking)                                                                                                                             |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>VF Management</b>                                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes VF                                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Clears before analyze and shock                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after shocks                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate airway management                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate cycles of drug-rhythm check/shock-CPR                                                                                                                                   |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Asystole Management</b>                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes asystole                                                                                                                                                                 |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Verbalizes potential reversible causes of asystole (H's and T's)                                                                                                                    |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after rhythm and pulse checks                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Post-Cardiac Arrest Care</b>                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Identifies ROSC                                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures BP and 12-lead ECG are performed, O <sub>2</sub> saturation is monitored, verbalizes need for endotracheal intubation and waveform capnography, and orders laboratory tests |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Considers targeted temperature management                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |

### STOP TEST

|                                                                                                                                                                                       |                                                                        |             |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------|-----------|
| <b>Test Results</b>                                                                                                                                                                   | Circle <b>PASS</b> or <b>NR</b> to indicate pass or needs remediation: | <b>PASS</b> | <b>NR</b> |
| Instructor Initials _____ Instructor Number _____ Date _____                                                                                                                          |                                                                        |             |           |
| <b>Learning Station Competency</b>                                                                                                                                                    |                                                                        |             |           |
| <input type="checkbox"/> Bradycardia <input type="checkbox"/> Tachycardia <input type="checkbox"/> Cardiac Arrest/Post-Cardiac Arrest Care <input type="checkbox"/> Megacode Practice |                                                                        |             |           |

# Megacode Practice Learning Station Checklist: Case 49/52/57/60/62

## Tachycardia → VF → PEA → PCAC

Student Name \_\_\_\_\_ Date of Test \_\_\_\_\_

| Critical Performance Steps                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    | Check if done correctly |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|-------------------------|
| <b>Team Leader</b>                                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Assigns team member roles                                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures high-quality CPR at all times                                                                                                                                               | Compression rate 100-120/min<br><input type="checkbox"/> | Compression depth of ≥2 inches<br><input type="checkbox"/> | Chest compression fraction >80%<br><input type="checkbox"/> | Chest recoil (optional)<br><input type="checkbox"/> | Ventilation (optional)<br><input type="checkbox"/> |                         |
| Ensures that team members communicate well                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Tachycardia Management</b>                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Starts oxygen if needed, places monitor, starts IV                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Places monitor leads in proper position                                                                                                                                             |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes unstable tachycardia                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Performs immediate synchronized cardioversion                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>VF Management</b>                                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes VF                                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Clears before analyze and shock                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after shocks                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate airway management                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate cycles of drug-rhythm check/shock-CPR                                                                                                                                   |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>PEA Management</b>                                                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes PEA                                                                                                                                                                      |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Verbalizes potential reversible causes of PEA (H's and T's)                                                                                                                         |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after rhythm and pulse checks                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Post-Cardiac Arrest Care</b>                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Identifies ROSC                                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures BP and 12-lead ECG are performed, O <sub>2</sub> saturation is monitored, verbalizes need for endotracheal intubation and waveform capnography, and orders laboratory tests |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Considers targeted temperature management                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |

### STOP TEST

|                                                                                                                                                                                       |                                                                        |             |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------|-----------|
| <b>Test Results</b>                                                                                                                                                                   | Circle <b>PASS</b> or <b>NR</b> to indicate pass or needs remediation: | <b>PASS</b> | <b>NR</b> |
| Instructor Initials _____ Instructor Number _____ Date _____                                                                                                                          |                                                                        |             |           |
| <b>Learning Station Competency</b>                                                                                                                                                    |                                                                        |             |           |
| <input type="checkbox"/> Bradycardia <input type="checkbox"/> Tachycardia <input type="checkbox"/> Cardiac Arrest/Post-Cardiac Arrest Care <input type="checkbox"/> Megacode Practice |                                                                        |             |           |

# Megacode Practice Learning Station Checklist: Case 50

## Bradycardia → Pulseless VT → Asystole → PCAC

Student Name \_\_\_\_\_ Date of Test \_\_\_\_\_

| Critical Performance Steps                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    | Check if done correctly |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|-------------------------|
| <b>Team Leader</b>                                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Assigns team member roles                                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures high-quality CPR at all times                                                                                                                                               | Compression rate 100-120/min<br><input type="checkbox"/> | Compression depth of ≥2 inches<br><input type="checkbox"/> | Chest compression fraction >80%<br><input type="checkbox"/> | Chest recoil (optional)<br><input type="checkbox"/> | Ventilation (optional)<br><input type="checkbox"/> |                         |
| Ensures that team members communicate well                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Bradycardia Management</b>                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Starts oxygen if needed, places monitor, starts IV                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Places monitor leads in proper position                                                                                                                                             |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes symptomatic bradycardia                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers correct dose of atropine                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Prepares for second-line treatment                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Pulseless VT Management</b>                                                                                                                                                      |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes pVT                                                                                                                                                                      |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Clears before analyze and shock                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after shocks                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate airway management                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate cycles of drug-rhythm check/shock-CPR                                                                                                                                   |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Asystole Management</b>                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes asystole                                                                                                                                                                 |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Verbalizes potential reversible causes of asystole (H's and T's)                                                                                                                    |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after rhythm and pulse checks                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Post-Cardiac Arrest Care</b>                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Identifies ROSC                                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures BP and 12-lead ECG are performed, O <sub>2</sub> saturation is monitored, verbalizes need for endotracheal intubation and waveform capnography, and orders laboratory tests |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Considers targeted temperature management                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |

### STOP TEST

|                                                                                                                                                                                       |                                                                        |             |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------|-----------|
| <b>Test Results</b>                                                                                                                                                                   | Circle <b>PASS</b> or <b>NR</b> to indicate pass or needs remediation: | <b>PASS</b> | <b>NR</b> |
| Instructor Initials _____ Instructor Number _____ Date _____                                                                                                                          |                                                                        |             |           |
| <b>Learning Station Competency</b>                                                                                                                                                    |                                                                        |             |           |
| <input type="checkbox"/> Bradycardia <input type="checkbox"/> Tachycardia <input type="checkbox"/> Cardiac Arrest/Post-Cardiac Arrest Care <input type="checkbox"/> Megacode Practice |                                                                        |             |           |

# Megacode Practice Learning Station Checklist: Case 51/54

## Bradycardia → Pulseless VT → PEA → PCAC

Student Name \_\_\_\_\_ Date of Test \_\_\_\_\_

| Critical Performance Steps                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    | Check if done correctly |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|-------------------------|
| <b>Team Leader</b>                                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Assigns team member roles                                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures high-quality CPR at all times                                                                                                                                               | Compression rate 100-120/min<br><input type="checkbox"/> | Compression depth of ≥2 inches<br><input type="checkbox"/> | Chest compression fraction >80%<br><input type="checkbox"/> | Chest recoil (optional)<br><input type="checkbox"/> | Ventilation (optional)<br><input type="checkbox"/> |                         |
| Ensures that team members communicate well                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Bradycardia Management</b>                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Starts oxygen if needed, places monitor, starts IV                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Places monitor leads in proper position                                                                                                                                             |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes symptomatic bradycardia                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers correct dose of atropine                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Prepares for second-line treatment                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Pulseless VT Management</b>                                                                                                                                                      |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes pVT                                                                                                                                                                      |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Clears before analyze and shock                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after shocks                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate airway management                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate cycles of drug-rhythm check/shock-CPR                                                                                                                                   |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>PEA Management</b>                                                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes PEA                                                                                                                                                                      |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Verbalizes potential reversible causes of PEA (H's and T's)                                                                                                                         |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after rhythm and pulse checks                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Post-Cardiac Arrest Care</b>                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Identifies ROSC                                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures BP and 12-lead ECG are performed, O <sub>2</sub> saturation is monitored, verbalizes need for endotracheal intubation and waveform capnography, and orders laboratory tests |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Considers targeted temperature management                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |

### STOP TEST

|                                                                                                                                                                                       |                                                                        |             |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------|-----------|
| <b>Test Results</b>                                                                                                                                                                   | Circle <b>PASS</b> or <b>NR</b> to indicate pass or needs remediation: | <b>PASS</b> | <b>NR</b> |
| Instructor Initials _____ Instructor Number _____ Date _____                                                                                                                          |                                                                        |             |           |
| <b>Learning Station Competency</b>                                                                                                                                                    |                                                                        |             |           |
| <input type="checkbox"/> Bradycardia <input type="checkbox"/> Tachycardia <input type="checkbox"/> Cardiac Arrest/Post-Cardiac Arrest Care <input type="checkbox"/> Megacode Practice |                                                                        |             |           |

# Megacode Practice Learning Station Checklist: Case 53

## Tachycardia → VF → Asystole → PCAC

Student Name \_\_\_\_\_ Date of Test \_\_\_\_\_

| Critical Performance Steps                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    | Check if done correctly |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|-------------------------|
| <b>Team Leader</b>                                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Assigns team member roles                                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures high-quality CPR at all times                                                                                                                                               | Compression rate 100-120/min<br><input type="checkbox"/> | Compression depth of ≥2 inches<br><input type="checkbox"/> | Chest compression fraction >80%<br><input type="checkbox"/> | Chest recoil (optional)<br><input type="checkbox"/> | Ventilation (optional)<br><input type="checkbox"/> |                         |
| Ensures that team members communicate well                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Tachycardia Management</b>                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Starts oxygen if needed, places monitor, starts IV                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Places monitor leads in proper position                                                                                                                                             |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes unstable tachycardia                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes symptoms due to tachycardia                                                                                                                                              |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Performs immediate synchronized cardioversion                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>VF Management</b>                                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes VF                                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Clears before analyze and shock                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after shocks                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate airway management                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate cycles of drug-rhythm check/shock-CPR                                                                                                                                   |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Asystole Management</b>                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes asystole                                                                                                                                                                 |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Verbalizes potential reversible causes of asystole (H's and T's)                                                                                                                    |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after rhythm and pulse checks                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Post-Cardiac Arrest Care</b>                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Identifies ROSC                                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures BP and 12-lead ECG are performed, O <sub>2</sub> saturation is monitored, verbalizes need for endotracheal intubation and waveform capnography, and orders laboratory tests |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Considers targeted temperature management                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |

### STOP TEST

|                                                                                                                                                                                       |                                                                        |             |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------|-----------|
| <b>Test Results</b>                                                                                                                                                                   | Circle <b>PASS</b> or <b>NR</b> to indicate pass or needs remediation: | <b>PASS</b> | <b>NR</b> |
| Instructor Initials _____ Instructor Number _____ Date _____                                                                                                                          |                                                                        |             |           |
| <b>Learning Station Competency</b>                                                                                                                                                    |                                                                        |             |           |
| <input type="checkbox"/> Bradycardia <input type="checkbox"/> Tachycardia <input type="checkbox"/> Cardiac Arrest/Post-Cardiac Arrest Care <input type="checkbox"/> Megacode Practice |                                                                        |             |           |

# Megacode Practice Learning Station Checklist: Case 55/58

## Tachycardia → Pulseless VT → PEA → PCAC

Student Name \_\_\_\_\_ Date of Test \_\_\_\_\_

| Critical Performance Steps                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    | Check if done correctly |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|-------------------------|
| <b>Team Leader</b>                                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Assigns team member roles                                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures high-quality CPR at all times                                                                                                                                               | Compression rate 100-120/min<br><input type="checkbox"/> | Compression depth of ≥2 inches<br><input type="checkbox"/> | Chest compression fraction >80%<br><input type="checkbox"/> | Chest recoil (optional)<br><input type="checkbox"/> | Ventilation (optional)<br><input type="checkbox"/> |                         |
| Ensures that team members communicate well                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Tachycardia Management</b>                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Starts oxygen if needed, places monitor, starts IV                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Places monitor leads in proper position                                                                                                                                             |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes unstable tachycardia                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes symptoms due to tachycardia                                                                                                                                              |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Performs immediate synchronized cardioversion                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Pulseless VT Management</b>                                                                                                                                                      |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes pulseless VT                                                                                                                                                             |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Clears before analyze and shock                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after shocks                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate airway management                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate cycles of drug-rhythm check/shock-CPR                                                                                                                                   |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>PEA Management</b>                                                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes PEA                                                                                                                                                                      |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Verbalizes potential reversible causes of PEA (H's and T's)                                                                                                                         |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after rhythm and pulse checks                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Post-Cardiac Arrest Care</b>                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Identifies ROSC                                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures BP and 12-lead ECG are performed, O <sub>2</sub> saturation is monitored, verbalizes need for endotracheal intubation and waveform capnography, and orders laboratory tests |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Considers targeted temperature management                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |

### STOP TEST

|                                                                                                                                                                                       |                                                                        |             |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------|-----------|
| <b>Test Results</b>                                                                                                                                                                   | Circle <b>PASS</b> or <b>NR</b> to indicate pass or needs remediation: | <b>PASS</b> | <b>NR</b> |
| Instructor Initials _____ Instructor Number _____ Date _____                                                                                                                          |                                                                        |             |           |
| <b>Learning Station Competency</b>                                                                                                                                                    |                                                                        |             |           |
| <input type="checkbox"/> Bradycardia <input type="checkbox"/> Tachycardia <input type="checkbox"/> Cardiac Arrest/Post-Cardiac Arrest Care <input type="checkbox"/> Megacode Practice |                                                                        |             |           |

# Megacode Practice Learning Station Checklist: Case 56/59

## Bradycardia → VF → Asystole → PCAC

Student Name \_\_\_\_\_ Date of Test \_\_\_\_\_

| Critical Performance Steps                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    | Check if done correctly |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|-------------------------|
| <b>Team Leader</b>                                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Assigns team member roles                                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures high-quality CPR at all times                                                                                                                                               | Compression rate 100-120/min<br><input type="checkbox"/> | Compression depth of ≥2 inches<br><input type="checkbox"/> | Chest compression fraction >80%<br><input type="checkbox"/> | Chest recoil (optional)<br><input type="checkbox"/> | Ventilation (optional)<br><input type="checkbox"/> |                         |
| Ensures that team members communicate well                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Bradycardia Management</b>                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Starts oxygen if needed, places monitor, starts IV                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Places monitor leads in proper position                                                                                                                                             |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes symptomatic bradycardia                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers correct dose of atropine                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Prepares for second-line treatment                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>VF Management</b>                                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes VF                                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Clears before analyze and shock                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after shocks                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate airway management                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate cycles of drug-rhythm check/shock-CPR                                                                                                                                   |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Asystole Management</b>                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes asystole                                                                                                                                                                 |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Verbalizes potential reversible causes of asystole (H's and T's)                                                                                                                    |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after rhythm and pulse checks                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Post-Cardiac Arrest Care</b>                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Identifies ROSC                                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures BP and 12-lead ECG are performed, O <sub>2</sub> saturation is monitored, verbalizes need for endotracheal intubation and waveform capnography, and orders laboratory tests |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Considers targeted temperature management                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |

### STOP TEST

|                                                                                                                                                                                       |                                                                        |             |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------|-----------|
| <b>Test Results</b>                                                                                                                                                                   | Circle <b>PASS</b> or <b>NR</b> to indicate pass or needs remediation: | <b>PASS</b> | <b>NR</b> |
| Instructor Initials _____ Instructor Number _____ Date _____                                                                                                                          |                                                                        |             |           |
| <b>Learning Station Competency</b>                                                                                                                                                    |                                                                        |             |           |
| <input type="checkbox"/> Bradycardia <input type="checkbox"/> Tachycardia <input type="checkbox"/> Cardiac Arrest/Post-Cardiac Arrest Care <input type="checkbox"/> Megacode Practice |                                                                        |             |           |

# Megacode Practice Learning Station Checklist: Case 61

## Tachycardia → VF → PEA → PCAC

Student Name \_\_\_\_\_ Date of Test \_\_\_\_\_

| Critical Performance Steps                                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    | Check if done correctly |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|-------------------------|
| <b>Team Leader</b>                                                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Assigns team member roles                                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures high-quality CPR at all times                                                                                                                                               | Compression rate 100-120/min<br><input type="checkbox"/> | Compression depth of ≥2 inches<br><input type="checkbox"/> | Chest compression fraction >80%<br><input type="checkbox"/> | Chest recoil (optional)<br><input type="checkbox"/> | Ventilation (optional)<br><input type="checkbox"/> |                         |
| Ensures that team members communicate well                                                                                                                                          |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Tachycardia Management</b>                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Starts oxygen if needed, places monitor, starts IV                                                                                                                                  |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Places monitor leads in proper position                                                                                                                                             |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes unstable tachycardia                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes symptoms due to gunshot wound                                                                                                                                            |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>VF Management</b>                                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes VF                                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Clears before analyze and shock                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after shocks                                                                                                                                                |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate airway management                                                                                                                                                       |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Appropriate cycles of drug-rhythm check/shock-CPR                                                                                                                                   |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>PEA Management</b>                                                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Recognizes PEA                                                                                                                                                                      |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Verbalizes potential reversible causes of PEA (H's and T's)                                                                                                                         |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Administers appropriate drug(s) and doses                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Immediately resumes CPR after rhythm and pulse checks                                                                                                                               |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| <b>Post-Cardiac Arrest Care</b>                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Identifies ROSC                                                                                                                                                                     |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Ensures BP and 12-lead ECG are performed, O <sub>2</sub> saturation is monitored, verbalizes need for endotracheal intubation and waveform capnography, and orders laboratory tests |                                                          |                                                            |                                                             |                                                     |                                                    |                         |
| Considers targeted temperature management                                                                                                                                           |                                                          |                                                            |                                                             |                                                     |                                                    |                         |

### STOP TEST

|                                                                                                                                                                                       |                                                                        |             |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------|-----------|
| <b>Test Results</b>                                                                                                                                                                   | Circle <b>PASS</b> or <b>NR</b> to indicate pass or needs remediation: | <b>PASS</b> | <b>NR</b> |
| Instructor Initials _____ Instructor Number _____ Date _____                                                                                                                          |                                                                        |             |           |
| <b>Learning Station Competency</b>                                                                                                                                                    |                                                                        |             |           |
| <input type="checkbox"/> Bradycardia <input type="checkbox"/> Tachycardia <input type="checkbox"/> Cardiac Arrest/Post-Cardiac Arrest Care <input type="checkbox"/> Megacode Practice |                                                                        |             |           |